

The Geometry of Gravitational-Wave Inference: a lawful anatomy of posteriors, systematics, and a geometric test of general relativity

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Abstract

The parameter-estimation posteriors of compact-binary gravitational-wave events have a shape that is usually treated as a nuisance. We show instead that this shape is *lawful*: the principal-axis orientation of each source-frame (m_1, m_2) posterior is quantitatively predicted, with *no fitted parameters*, by the constant-chirp-mass curve evaluated at the event’s own median chirp mass over its mass-ratio marginal. The local constant- $\mathcal{M}_c$ tangent — the Fisher/metric approximation underlying rapid parameter-estimation tools — leaves residuals that grow with posterior elongation; the curve (chord) term removes them. Derived on GWTC-3 and locked before opening later catalogs, the law predicts the measured orientation out-of-sample on the disjoint GWTC-4.0/O4a and GWTC-5.0/O4b releases to a median mismatch of 1.26° and 1.22° respectively. A physics-blind search finds *no* source-class dependence of the residuals: precession, spin, mass ratio, redshift, mass and neutron-star–black-hole character all fail to correlate after multiple-comparison control; the only surviving correlate is waveform-model disagreement, a systematic. Inverting the law to *measure* the leading inspiral mass-combination exponent yields 0.628 ± 0.009 (stat) ± 0.016 (syst), consistent with the general-relativistic value $3/5$; a naive analysis produced a spurious 3σ “deviation” that a coherence and systematic-floor analysis correctly demotes to 1.5σ . We ground these results in the established mathematics of measurement (Fisher–Rao information geometry, sloppy models, principal curves, Backus–Gilbert resolution theory), from which the lawfulness is expected, and argue that the contribution is the predictive, catalog-validated *application* to gravitational-wave posteriors and its use as a systematics diagnostic and an independent test of gravity.

1 Introduction

Gravitational-wave (GW) parameter estimation delivers, for each detected compact-binary coalescence, a posterior probability distribution over the source parameters. It is a commonplace that the chirp mass $\mathcal{M}_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$ is the best-measured mass combination and that the component-mass posterior forms an elongated “banana” lying along near-constant- $\mathcal{M}_c$ contours [1; 2; 4]. This paper asks a sharper question: is the *geometry* of that posterior — specifically the orientation of its principal axis — quantitatively *predictable*, event by event, with no free parameters, and does the prediction survive out-of-sample across independent catalogs?

We answer yes. The organizing thesis is that the uncertainty geometry of GW inference is *measurement optics*: it is a property of what the instrument and signal can and cannot resolve, largely independent of source astrophysics. This viewpoint is not new mathematically — it is

the content of Fisher–Rao information geometry [7], the “sloppy model” framework [8], principal curves [9], and Backus–Gilbert resolution theory [10]. Our contribution is to make it *predictive and validated* for GW posteriors, and to turn it into two tools: a systematics detector and an independent, waveform-fit-free test of general relativity (GR).

2 The curved chirp-mass law

2.1 Definition

For an event with source-frame samples $\{(m_1, m_2)\}$, let ψ_{meas} be the position angle of the principal axis of the sample covariance, and let the axis ratio axr be the ratio of its principal standard deviations. The constant- $\mathcal{M}_c$ *curve* through the plane is

$$m_1(q) = \mathcal{M}_c (1 + q)^{1/5} q^{-3/5}, \quad m_2 = q m_1, \quad (1)$$

where $q = m_2/m_1 \in (0, 1]$ and $\mathcal{M}_c$ is the event’s median source-frame chirp mass. The predicted orientation ψ_{curve} is the principal axis of this curve evaluated over the event’s own q marginal. The local linear approximation ψ_{tangent} is the tangent to the contour at the median point. There are no calibration constants: every ingredient is supplied by the event.

2.2 Out-of-sample validation across three catalogs

The law was derived from GWTC-1/2.1/3 events and its prediction placed on record before any later-catalog file was opened. Table 1 reports the locked decision on two subsequent, disjoint catalogs. On elongated events ($\text{axr} \geq 3$, where orientation is well defined) the median mismatch $|\psi_{\text{curve}} - \psi_{\text{meas}}|$ is 1.26° (GWTC-4.0/O4a) and 1.22° (GWTC-5.0/O4b); the curve beats the tangent on the full catalogs, and the tangent-law replication statistics reproduce the training-set values. The catalogs are mutually disjoint (zero shared events).

	GWTC-3 (train)	GWTC-4.0/O4a	GWTC-5.0/O4b
median $ \psi_{\text{curve}} - \psi_{\text{meas}} $ (elongated)	—	1.26°	1.22°
median $ \psi_{\text{curve}} - \psi_{\text{meas}} $ (all)	—	7.53°	5.41°
median $ \psi_{\text{tangent}} - \psi_{\text{meas}} $ (all)	7.49°	9.36°	8.23°
chirp-vs-total win fraction	81%	86%	88%
Spearman($ \Delta\psi_{\text{tangent}} $, axr)	−0.42	−0.40	−0.69
N events (elongated)	—	86 (19)	104 (32)

Table 1: The curved chirp-mass law, locked before data and scored out-of-sample. The curve prediction lands within $\sim 1.2^\circ$ of the measured orientation on elongated events across two independent catalogs; the tangent leaves a systematic residual that grows with elongation (negative Spearman).

3 Why it is lawful: the mathematics of measurement

The law is a consequence, not a coincidence. The GW inspiral phase is dominated at leading (0PN) order by $\mathcal{M}_c$, so the Fisher information has one sharply-measured (“stiff”) direction — constant $\mathcal{M}_c$ — and poorly-measured (“sloppy”) directions transverse to it [8]. By the Bernstein–von Mises theorem the posterior tends to a Gaussian with covariance equal to the inverse Fisher matrix [7], so its long axis aligns with the constant- $\mathcal{M}_c$ level set; by Čencov’s theorem the Fisher–Rao metric

is the unique metric invariant under sufficient statistics, and $\mathcal{M}_c$ is approximately sufficient for the inspiral. The distinction between the tangent and the curve is precisely the distinction between linear principal-component analysis and a *principal curve* [9]: a chord of a curved level set deviates from its tangent by an amount that grows with the chord length, which is the observed elongation dependence. The “resolved vs. unresolved directions” of Backus–Gilbert resolution theory [10] are the same split in the inverse-problem language. To our knowledge, none of this literature predicts the per-event (m_1, m_2) posterior orientation from the chirp-mass curve, decomposes the tangent-vs-curve residual against elongation, or validates such a law out-of-sample across catalogs; that predictive application is the contribution here.

4 Coherence as a systematics detector

If the geometry is measurement optics, then *departures* from it are informative, and departures that are *coherent* across events are systematics rather than physics. We test this in two ways. First, a physics-blind “outlier atlas”: ranking the elongated O4b events by their curve residual and correlating against nine pre-declared axes (precession χ_p , aligned spin χ_{eff} , mass ratio, redshift, signal-to-noise ratio, chirp mass, total mass, secondary mass, and waveform-family disagreement) with Benjamini–Hochberg control and a partial correlation removing the axis-ratio confound, *no physical axis* survives; the only surviving correlate is waveform-model disagreement — a measurement systematic, with a sign indicating that the curve tracks the waveform *ensemble* consensus better than any single waveform tracks another. Second, and more sharply, the geometric GR-exponent measurement of Section 5 exhibits a coherent 3σ “deviation” that the same lens correctly identifies as a systematic. The methodological rule — an anomaly coherent across independent datasets is a systematic, not a signal — is thus both stated and exercised.

5 A geometric test of general relativity

The exponents of $\mathcal{M}_c$ are fixed by the 0PN quadrupole formula. Generalize to the one-parameter family

$$\mathcal{M}_p = (m_1 m_2)^p (m_1 + m_2)^{1-2p}, \quad \text{GR: } p = 3/5 = 0.600, \quad (2)$$

and *measure* p from the posterior orientations: because orientation is scale-free ($m_1(q; p) \propto q^{-p}(1+q)^{2p-1}$), the fit isolates the exponent from the mass scale. A deviation would signal modified inspiral phasing (e.g. dipole radiation). On the O4b elongated events alone we find $p = 0.616 \pm 0.011$; combining O4a and O4b, $p = 0.628 \pm 0.009$.

Taken naively this is a coherent 3.1σ departure from GR — and it is a false alarm. Two pre-committed checks demote it. (i) The two independent catalogs disagree with each other by 0.031, as large as the offset from GR (0.028); this catalog-to-catalog spread is a systematic that the bootstrap statistical error (0.009) does not capture. Including it, $p = 0.628 \pm 0.009$ (stat) ± 0.016 (syst), or 1.5σ from GR. (ii) The per-event exponent depends on elongation (Spearman = -0.41 , coherent in both catalogs); the cleanest, most-elongated half yields $p = 0.606$, consistent with GR. The offset is therefore a finite-width imperfection of the leading-order geometric model, coherent across catalogs, and vanishing in the clean limit — not a violation of GR. We stress that this test is a methodologically *independent* cross-check, not more sensitive than direct parameterized-deviation bounds; its value is that it shares none of the waveform-template machinery.

6 The information anatomy of a detection

The same Fisher structure that fixes the posterior orientation also fixes *when*, in frequency, each parameter’s information arrives. Computing the cumulative Fisher information for the chirp mass (0PN), mass ratio (1PN) and spin (1.5PN) across all 190 real O4a+O4b events, the ordering $\mathcal{M}_c < \eta < \chi$ holds universally: the chirp mass is always determined earliest (~ 35 Hz), spin latest, and the frequency separation — the anatomy’s “richness” — is a monotonic function of total mass (Spearman -1.00), widest for light, long-inspiral systems and compressed for heavy ones that merge immediately in band. This is a frequency-resolved, per-event map of what each detection measured and when. It is a forward prediction from the waveform model; a fully data-driven version (re-analysing loud events in frequency slices and comparing empirical to GR-predicted information accumulation) is left to future work.

7 The loudest event: GW250114

As an application, GW250114 — the loudest event to date (network SNR ~ 80), a near-equal-mass binary black hole with remnant $M_f^{\text{src}} \approx 62.7 M_\odot$, $\chi_f \approx 0.68$ — is essentially a much louder counterpart of the first detection GW150914. Its remnant predicts Kerr quasi-normal-mode frequencies of 248.9 Hz ($\tau = 4.10$ ms) for the fundamental and 243.4 Hz ($\tau = 1.35$ ms) for the first overtone; published ringdown spectroscopy detects the overtone at 99.9% confidence and finds no deviation from the Kerr prediction [11], confirming the no-hair theorem at record signal-to-noise. Being near-equal-mass, its posterior is nearly round and it is (by construction) outside the elongated set used for the curved law — an illustration that the loudest events are not always the most informative for a given geometric probe.

8 Discussion

We have shown that the mass-plane posterior geometry of GW events obeys a predictive, zero-parameter constant-chirp-mass curve law, validated out-of-sample across three catalogs; that its residuals are physics-blind (a systematics detector, not a source-class probe); and that inverting it yields an independent geometric test of GR’s inspiral mass structure, consistent with general relativity once systematics are controlled. The results rest on well-understood measurement mathematics, which is a strength: the lawfulness is expected, and the value lies in the predictive application, the out-of-sample discipline, and the diagnostic uses.

Honest limitations. (1) The curved law is validated on elongated events; round posteriors have ill-defined orientation and are excluded by design. (2) The geometric GR test is a cross-check, less sensitive than dedicated waveform-based bounds, and is currently systematics-limited by a finite-width model imperfection. (3) The information anatomy is model-predicted, not strain-extracted. (4) The first-principles derivation of the curve’s exact width and curvature from the Fisher information remains open. (5) Waveform-model choice contributes a per-event orientation scatter of a few degrees; it is incoherent and averages out of population-level quantities, but is the dominant per-event systematic.

Data availability. All data are public: GWTC-4.0 PE (Zenodo 16053484) and GWTC-5.0 PE (Zenodo 20276106, 20348006). Analysis code, locked pre-registrations, per-event results, and a reproducibility notebook accompany this work.

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